

Sivan Syed



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EDUCATION

The University of Texas at Austin (GPA: 3.8/4.0)

Bachelor of Science, **Electrical and Computer Engineering**

Certificate, **German**

May 2027

Bachelor of Science, **Physics**

Minor, **Materials Science and Engineering**

Relevant Coursework: Embedded Systems, Solid-State Devices, Power Electronics, Digital Logic Design, Fundamental Electronic Circuits

Graduate Coursework: Magnetic Materials and Devices, Computational Physics, Quantum Mechanics I

EXPERIENCE

Adom Industries Inc. / R&D Design Engineer / Electrical and Robotics Engineering

May 2025 – August 2025

- Architected a **48V autonomous robot's** Vehicle Control Unit (VCU), integrating an RP5 and RP2040 for **ROS path planning** with via **CAN bus**; optimized chassis topology via FEA, reducing material usage by 88%
- Engineered a **high-voltage power distribution network** featuring a **hardware-in-the-loop** E-stop circuit and heavy-duty contactor, preserving isolated 5V logic power for data logging
- Designed **custom PCBs** for **LiDAR** and **motor control**, applying star grounding and **digital isolators** to mitigate 48V EMI.

Sandia National Laboratories / R&D Computational Engineer / Pulsed Power Physics

May 2024 – August 2024

- Analyzed fluid **RTIs** and **RMIs** to diagnose inaccuracies in **ALEGRA simulations**
- Built a fast **current-pulse generator** to synthesize 10k+ waveforms for **ML-aided surrogate modeling** and **UQ pipelines**
- Wrote **ODE/trajectory solvers** to track radial liner implosions and compare ALEGRA predictions and observed instability spectra

FSAE Longhorn Racing / Dynamics + Vehicle Modeling Engineer / Steering Lead, Member

September 2022 – August 2025

- Designed an anti-Ackermann geometry using a system-level **Model-in-the-Loop (MIL)** Python **vehicle simulation**; validated **slip-angle** targets by tuning **compliance coefficients** and correlating simulation outputs against **physical track telemetry**
- Re-engineered steering column using **carbon-fiber** layups, cutting system weight $\approx 68\%$ while maintaining torsional rigidity
- Created a modular **gearbox CAD stack (GD&T, CNC and composite fab)** with fast-swap adjustability for testing days

2D Material Synthesis & Characterization – Banerjee Lab / Nanomaterials and Devices

September 2025 – Present

- Executed **Chemical Vapor Deposition (CVD)** processes to grow high-quality Molybdenum Disulfide (MoS_2) monolayers, **optimizing growth parameters** to control crystal orientation and thickness.
- Characterized semiconductor properties including conductivity and p/n-type doping through **XRD**, **Raman**, and **Photoluminescence (PL) spectroscopy** to validate material lattice structure and optical bandgaps.

Projects

IoT Smart Object – Wi-Fi Enabled Alarm Clock

Relevant Skills: Embedded Systems, C, Firmware, UART, SPI, MQTT, Network Protocol

- Interfaced a TM4C123 LaunchPad with an **ESP8266** Wi-Fi module via UART to enable bi-directional communication with a **web-based dashboard**
- Implemented a **MQTT publish-subscribe architecture** to remotely control device states and **synchronize real-time clock data** between the embedded hardware and a JavaScript-based web application

Ping Pong Ball Equilibrium Robot

Relevant Skills: Mechatronics, Robotics, Controls, C, C++, Power Electronics

- Developing inverse kinematics and PID control** of a **robotic platform** for precise movement correction within ± 0.5 mm accuracy.
- Integrating **high-torque servos**, **power circuits**, and custom **3D-printed components** to engineer the **electromechanical system**

E-Bike Boost Converter Design

Relevant Skills: Power Electronics, Embedded Systems, PCB Design, KiCAD, Circuit Design

- Engineered a custom PCB for a **high-power e-bike DC-DC boost converter**, mitigating radiated EMI and voltage spikes by minimizing the high di/dt **critical loop** and gate-drive parasitic inductance
- Architected a **digital control loop** on a **TI C2000** MCU, synchronizing the **ePWM module** with the ADC via hardware triggers to sample feedback without switching noise for **precise duty-cycle regulation**.

Attenuation Lengths of Scintillating Fibers

Relevant Skills: Python, SOLIDWORKS, Manufacturing, Sensor Integration, Data Analysis

- Measured and **analyzed attenuation lengths** of >100 wavelength shifting fibers from Eljen with 520nm wavelength LEDs
- Designed, **3D printed**, and **milled** mechanical mounts for a sensor system of **SiPMs**, **PMTs** and **Spectrometers**

Steering Gearbox

Relevant Skills: CAD, SOLIDWORKS, ANSYS, Manufacturing, CNC, Manual Mill, Lathe

- Performed **stress analyses** of various parts with **finite element analysis** from **Ansys** and **SOLIDWORKS**
- Calculated steering efforts and ratios with high level **vehicle modeling** and **computational verification**

SKILLS

Software Language Proficiencies: C, C++, ARM, Python, MATLAB, Bash, JS, HTML, CSS, LaTeX, Rust

Simulation/Modeling Software: KiCAD, Fusion360, LTSpice, FLASH, MaGe, SOLIDWORKS, VCarve, Inventor, Ansys